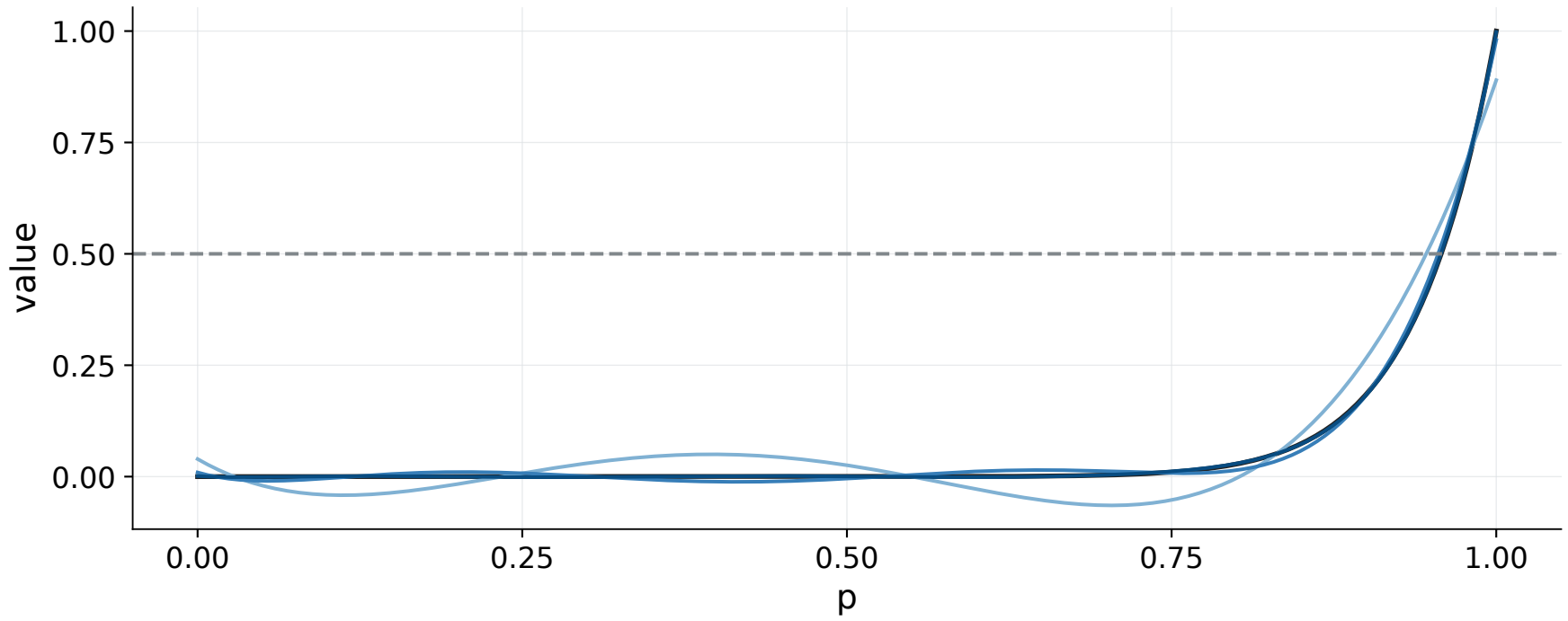
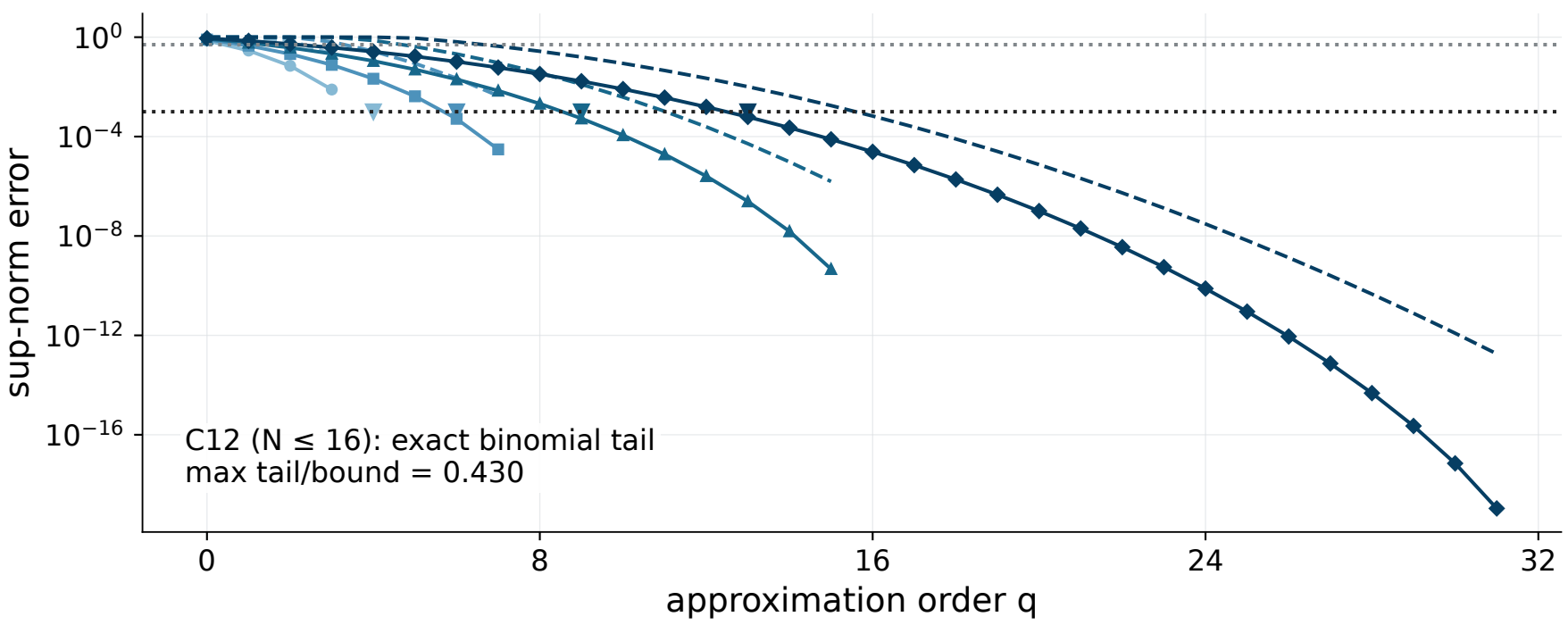
(a) $p^{16} \approx$ order-8 polynomial



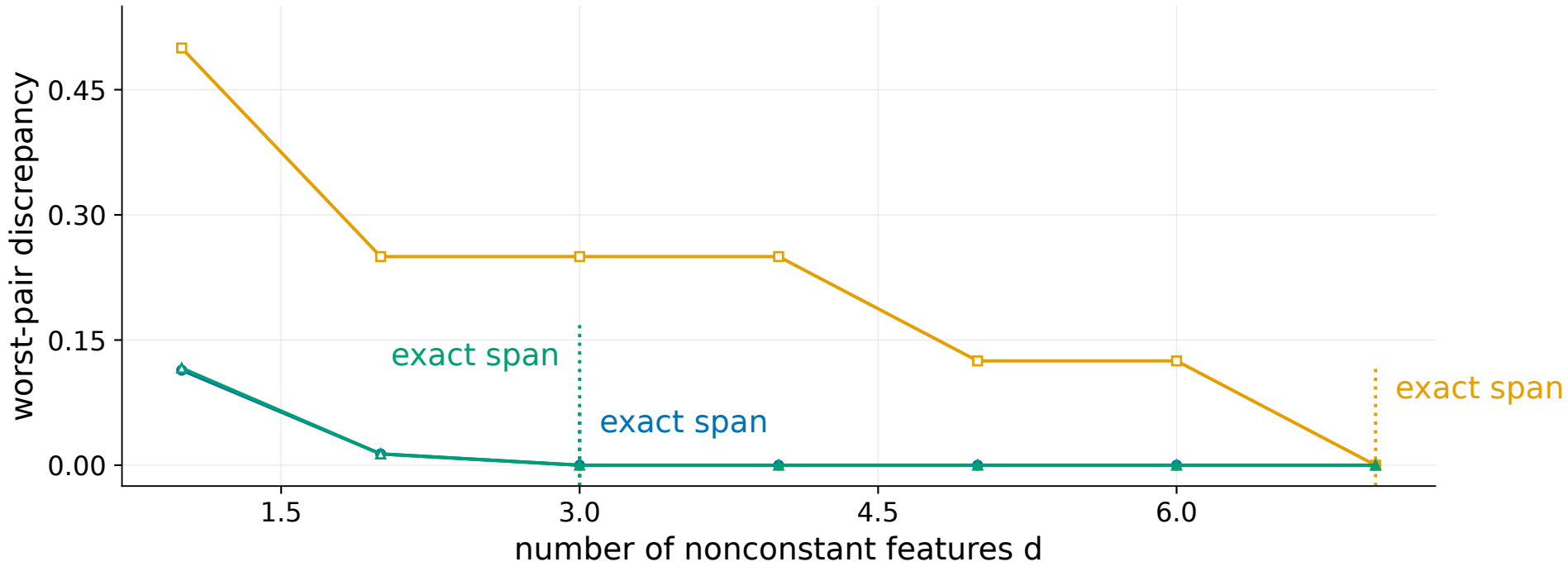
p^{16}
h_4, sup error 0.11
h_6, sup error 0.0201
h_8, sup error 0.0021
parity below degree N: constant 0.5 (best error 0.5)

(b) Needed order grows like √N



N=4
N=8
N=16
N=32
parity (all N)
threshold 0.001
Hoeffding upper
first crossing

(c) Parity needs all $2^3 - 1$ directions



scalar affine powers
parity pullbacks
structured sum powers

Exact finite computations.